Graphic Era University

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

GridRescue: Interactive Grid Failure and Load Redistribution Simulator

Team Name: *Team #*	*GridRescuers* *DSCPP-III-2026-T091*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

GridRescue is a C++ program that shows what can happen when a power line breaks. The grid is shown as points and connections. A point can be a power source or a place that uses electricity, such as a home or hospital. When a connection fails, the program aims to find which places lose their link to power and whether another route is available. If power is limited, important places should be served first. The project helps us use data structures and object-oriented programming to solve a problem based on real life. It is an educational model and does not control a real power grid.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

The project uses C++17 and is stored on GitHub. The grid is represented as a graph: points are nodes and power lines are edges. The repository README lists an adjacency list, BFS, DFS, Dijkstra's algorithm, a priority queue/min-heap, vectors, and an unordered map as planned or used concepts. Classes and objects help keep grid information and actions organised. Dear ImGui is listed for the graphical interface. The intended flow is to create the grid, add connections, simulate a failed line, find affected places, search for another route, give priority to important users, and share the available power. The repository describes the project as under development, so each algorithm and the interface still need to be checked together. No live power-company data or live grid connection is used.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Project purpose, team details, and the main plan were written in the Phase-I proposal.	All team members
A GitHub repository and README were created to describe the project, tools, and planned working steps.	All team members
The main data structures and algorithms for the grid have been identified in the project documentation.	All team members

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

One challenge is making all parts work together. A failed line must update the grid, the program must find the affected places, and the route search must use the updated connections. Another challenge is showing these changes clearly on the screen. We will solve these problems by testing one small example at a time, checking the results by hand, and fixing errors before testing larger examples. The README says the project is still under development, so we cannot say that every feature is complete or has passed testing yet.

Tasks Pending (7 pts)

(Describe the main tasks that you still need to complete. Max 250 words).

Task Pending	Team Member (to complete the task)
Finish and check the grid model, connection failure handling, and affected-node search.	GridRescuers team
Connect and verify alternate-route search and priority-based power sharing.	GridRescuers team
Complete the GUI, test common and error cases, and capture final results for the report.	GridRescuers team

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

The planned deliverable is a C++17 simulation of a small power grid with a simple graphical interface. It should let a user view the grid, simulate a broken connection, see which places are affected, and check whether another route can be used. It should also show how limited power can be given to important users first. The final submission will include the source code, a tested demonstration, and a report that explains the main steps in simple language. These are target outcomes; the repository currently marks the project as under development.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

The project idea, team information, planned algorithms, tools, and main workflow are documented. A GitHub repository contains the project README and an Implementation folder. The repository status says the project is under development and that the grid, algorithms, and simulation features are being built step by step. Because the available project information does not show a complete test report, the final working status of each feature is not confirmed here. The main work still to check is algorithm integration, GUI behaviour, power-priority handling, and full testing.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

Repository: `https://github.com/tejasvsinghnagarkoti-234/Gridrescue`
Main branch shown on GitHub: `main`.
The repository contains an `Implementation` folder, a `PHASE -1` folder, and a README. The README describes the project purpose, C++17, Dear ImGui, Git/GitHub, the planned algorithms, and future improvements. GitHub showed 68 commits when this report was prepared. Individual commit messages and the exact status of each source file could not be confirmed from the available repository view, so no specific commit is claimed as a completed feature.

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	**Status (Pass/Fail)**	**Notes**
Repository and documentation review	Reviewed	README and folder list checked. This is not a software test.
Algorithm and GUI tests	Not confirmed	No software test results were available to confirm.
Failure and alternate-route scenarios	Pending verification	Test after the related features are connected.

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

- Project plan and team details: completed in Phase I.
- GitHub repository and README: available.
- Grid model and failure simulation: in progress; final status needs testing.
- Alternate route and priority-based power sharing: pending verification.
- GUI, full testing, and final demonstration: pending verification.
- Phase-II report: prepared for review.